

Backend Endpoint Request Template

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TBF School Hub API Specification (v1) • Tanzania Baraka Foundation

1. Feature Name: Fetch Learning Materials

Purpose: Students and teachers need to view and download learning materials filtered by subject, form level, and category.

Frontend Page/Module: Student Portal !' Learning Materials / Teacher Portal !' Materials Library

Requested Operation: [X] Fetch Data (GET /api/v1/materials) [] Create [] Update [] Delete [] Upload File

Required Inputs:

Field	Type	Required	Description
resource_name	string	No	Subject name or search query (e.g., 'biology')
form_num	integer	No	Student form level (1 to 6)
category	string	No	Category: 'notes', 'past_papers', 'summaries', 'all'
page	integer	No	Page number (default: 1)
limit	integer	No	Page size (default: 20)

Example Request Query:

```
GET /api/v1/materials?resource_name=biology&form_num=1&category=notes&page=1&limit=20
```

Expected Response Data:

```
{
  "materials": [
    {
      "id": "mat-9d2a-481b-a48e",
      "title": "Biology Form One Notes",
      "subject": "Biology",
      "form_num": 1,
      "category": "notes",
      "resource_url": "https://storage.thebarakafoundation.or.tz/materials/bio_f1_notes.pdf",
      "created_at": "2026-08-31T10:00:00Z"
    }
  ],
  "total": 1,
  "page": 1,
  "total_pages": 1
}
```

Filters Needed: Filter by subject, category, form level, and school

Sorting Needed: Newest First (created_at DESC), Alphabetical (title ASC)

Pagination Needed: Yes (page, limit, total)

Authentication Required: Yes (Bearer Token) • Allowed Roles: Student, Teacher, School Admin

Error Cases: 401 Unauthorized, 404 No materials found, 422 Invalid form number

Additional Notes: Only return materials belonging to the student's assigned school or public TBF library.

2. Feature Name: AI Study Companion (Ask Baraka)

Purpose: Real-time, syllabus-grounded explanations and derivations for NECTA curriculum in English and Kiswahili.

Frontend Page/Module: Student Portal !' Ask Baraka AI Tutor / Teacher Portal !' AI Assistant

Requested Operation: [X] Create Data / Process (POST /api/v1/ai/ask) [] Fetch [] Upload

Example Request Data:

```
{
  "message": "Explain the difference between Light-dependent and Dark reactions in Photosynthesis.",
  "history": [
    {
      "text": "Habari Baraka!",
      "sender": "user"
    },
    {
      "text": "Habari! How can I help with your studies today?",
      "sender": "bot"
    }
  ],
  "subject_context": "Biology"
}
```

Expected Response Data:

```
{
  "text": "## Photosynthesis Reactions\n1. Light Reactions (Thylakoid): Photolysis of water yields ATP/NADPH.\n2. Calvin Cycle (Stroma): CO2 fixed into Glucose.",
  "source": "tbf-gemini-ai",
  "knowledge_sources": [
    "NECTA Form 1 Biology Syllabus",
    "TIE Secondary Science"
  ],
  "suggested_questions": [
    "What is photolysis of water?"
  ]
}
```

Authentication Required: Yes (Bearer Token) • Allowed Roles: Student, Solo Learner, Teacher

Error Cases: 400 Bad Request (empty message), 429 Rate Limit Exceeded

3. Feature Name: Auto-Arrange Material & AI Quiz Generator

Purpose: Teachers upload or paste raw lesson notes to auto-generate structured notes, summaries, and quiz questions.

Frontend Page/Module: Teacher Dashboard !' Material Creator / Admin Dashboard !' Curriculum Builder

Requested Operation: [X] Create Data / AI Process (POST /api/v1/arrange-material)

Example Request Data:

```
{
  "title": "Ohm's Law and Electrical Resistance",
  "subject": "Physics",
  "templateType": "notes",
  "rawText": "Ohm's law states that current through a conductor is directly proportional to voltage across it. Formula: V = I * R."
}
```

Expected Response Data:

```
{
  "arrangedContent": {
    "introduction": "Ohm's Law is a foundational principle of electrical physics.",
    "sections": [
      {
        "heading": "Formulation",
        "body": "V = I * R"
      }
    ],
    "keyPoints": [
      "Current is proportional to Voltage",
      "Resistance opposes current flow"
    ],
    "summary": "Key formula: V = I * R"
  }
}
```

```
},
"quizQuestions": [
  {
    "id": "q-1",
    "question": "What is the formula for Ohm's Law?",
    "options": [
      "V = I * R",
      "P = V * I",
      "I = V * R",
      "R = V * I"
    ],
    "correctAnswer": 0,
    "explanation": "Voltage equals Current multiplied by Resistance (V = I * R)."
  }
]
}
```

Authentication Required: Yes • Allowed Roles: Teacher, School Admin

4. Feature Name: School Registration & Verification

Purpose: New schools register institution, submit admin credentials, and upload MoE/NECTA certificates for approval.

Frontend Page/Module: Welcome Screen !' Register Your School (Steps 1–3)

Requested Operation: [X] Create Data (POST /api/v1/auth/register/school) [X] Upload File (POST /api/v1/auth/upload-verification-doc)

Example Request Data:

```
{
  "fname": "Paulo",
  "lname": "Michael",
  "email": "headmaster@barakaschool.ac.tz",
  "password": "SecurePassword2026!",
  "school_name": "Baraka Secondary School",
  "registration_number": "MoE/SEC/2026/0921",
  "region": "Dar es Salaam",
  "school_type": "Secondary",
  "headmaster_name": "Paulo Michael Senior",
  "verification_document_name": "moe_registration_cert.pdf"
}
```

Expected Response Data:

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "expires_in": 86400,
  "user": {
    "id": "usr_94a7e2b1",
    "email": "headmaster@barakaschool.ac.tz",
    "role": "school_admin"
  },
  "school": {
    "id": "sch_18f4a01c",
    "name": "Baraka Secondary School",
    "verification_status": "pending_review"
  }
}
```

Authentication Required: No (Public Onboarding)

Error Cases: 409 Conflict (Email/Reg No exists), 422 Unprocessable (Missing document)

5. Feature Name: Student & Solo Learner Registration

Purpose: Enrolls individual students / solo learners with grade level and curriculum choice.

Requested Operation: [X] Create Data (POST /api/v1/auth/register/student)

```
{
  "fname": "Amani",
  "lname": "Baraka",
  "email": "amani@student.ac.tz",
  "study_level": "Form 1",
  "curriculum": "Tanzania National (NECTA)"
}
```

6. Feature Name: Quizzes & Assessment Submissions

Purpose: Submit quiz test answers to calculate score, award XP, and track topic mastery.

Requested Operation: [X] Fetch (GET /api/v1/quizzes) [X] Create (POST /api/v1/quizzes/submit)

```
{
  "quiz_id": "quiz_bio_f1_01",
  "student_id": "usr_88b12e",
  "score": 3,
  "total": 3,
  "answers": [
    {
      "question_id": "q-1",
      "selected": 0,
      "is_correct": true
    }
  ]
}
```


7. School Management CRUD Summary Table

Operation	Method & Route	Allowed Roles	Purpose
Fetch Students	GET /api/v1/students	Admin, Teacher	List students with class filter & search
Create Student	POST /api/v1/students	Admin	Register new student in school
Update Student	PUT /api/v1/students/:id	Admin	Modify student class/details
Delete Student	DELETE /api/v1/students/:id	Admin	Remove student record
Fetch Teachers	GET /api/v1/teachers	Admin	List teachers with subjects & classes
Create Teacher	POST /api/v1/teachers	Admin	Add teacher and assign credentials
Fetch Timetable	GET /api/v1/timetable	Admin, Teacher, Student	Get daily schedule & assigned rooms
Create Slot	POST /api/v1/timetable	Admin	Schedule new timetable period
Accept Slot	PUT /api/v1/timetable/:id/accept	Teacher	Teacher confirms timetable slot

*All endpoints conform to TBF
School Hub security standards
with JSON payload over
HTTPS. Base URL: [https://schub
api.thebarakafoundation.or.tz/
api/v1](https://schub
api.thebarakafoundation.or.tz/
api/v1)*